

CO-1

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Assessment Tool-1

- (1) The company wants to automatic its payroll process
Write a c program to calculate the total salary by
adding the basic salary and bonus amount of an
employee.

Pseudo Code

Start

Read Basic Salary, Bonus

Total Salary = Basic Salary + Bonus

Print Total Salary

Stop

Input :

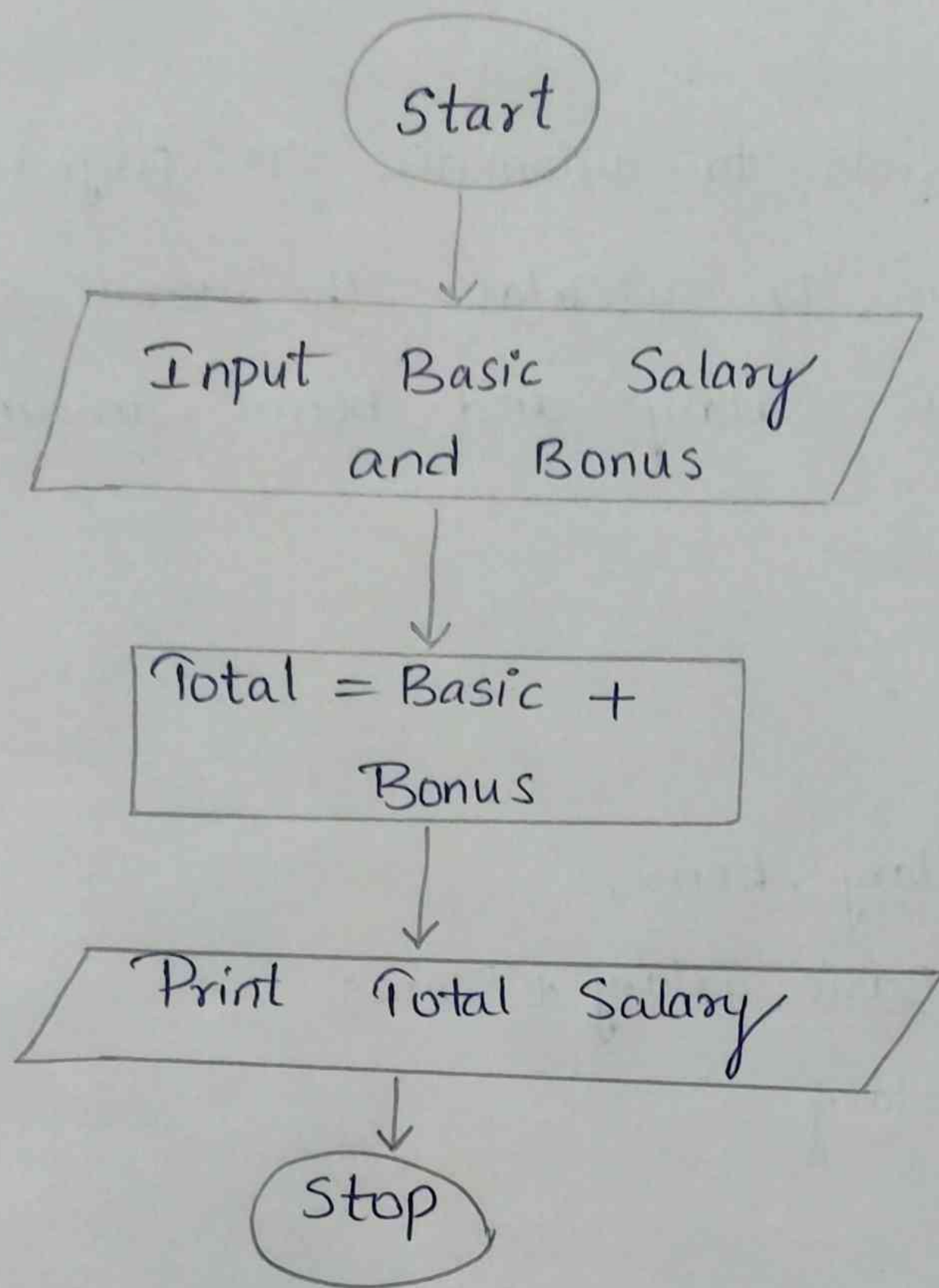
Basic Salary

Bonus

Output :

Total Salary

Flowchart:



2. During a database update, two employee ID's need to be exchanged. Write a C program to swap two numbers using a temporary variable.

Pseudo code:

Read

Start

Read A, B

Temp = A

A = B

B = Temp

Print A, B

STOP

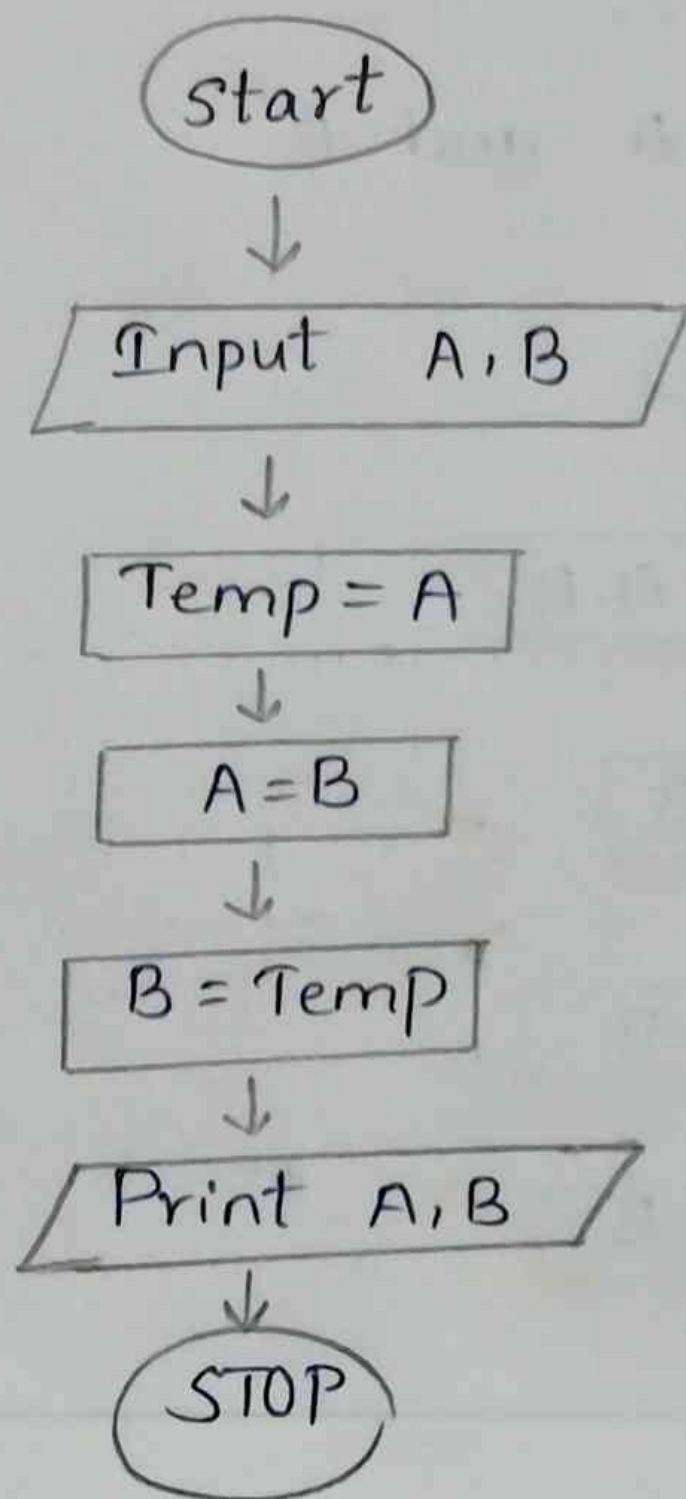
Input:

- A
- B

Output:

Swapped values of A and B

Flow chart:



3. In an embedded system with limited memory two variables must be exchanged without using any extra storage. Write a C program to swap two numbers without using two numbers.

Pseudo code:

START

Read A, B

$A = A + B$

$B = A - B$

$A = A - B$

Print A, B

STOP

Input:

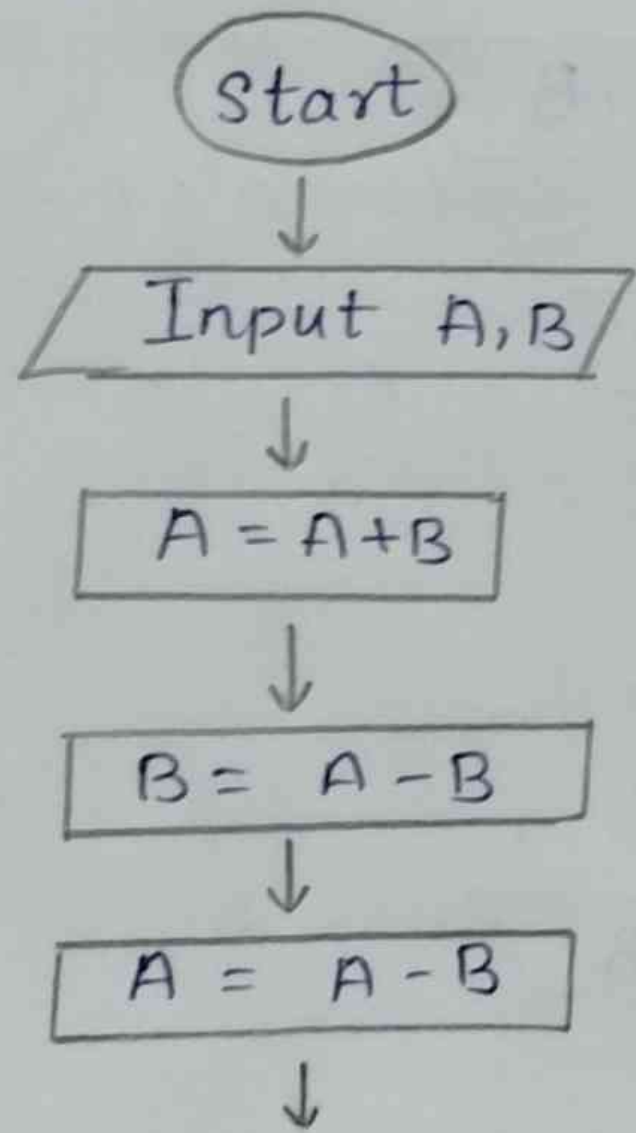
A

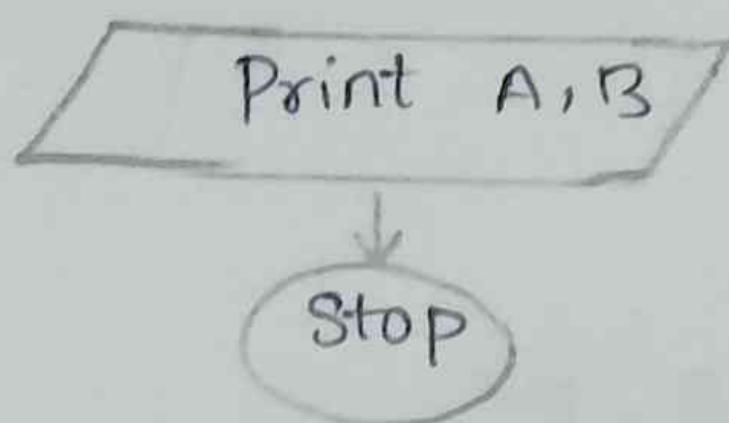
B

Output:

Swapped values of A and B

Flowchart:





4. A bank wants to provide customers with an estimate of the interest earned on their deposit write a C program to calculate Simple interest using the formula.

Pseudo code:

Start

Read Principal, Rate, Time

$SI = (Principal \times Rate \times Time) / 100$

Print SI

STOP

Input:

Principal (P)

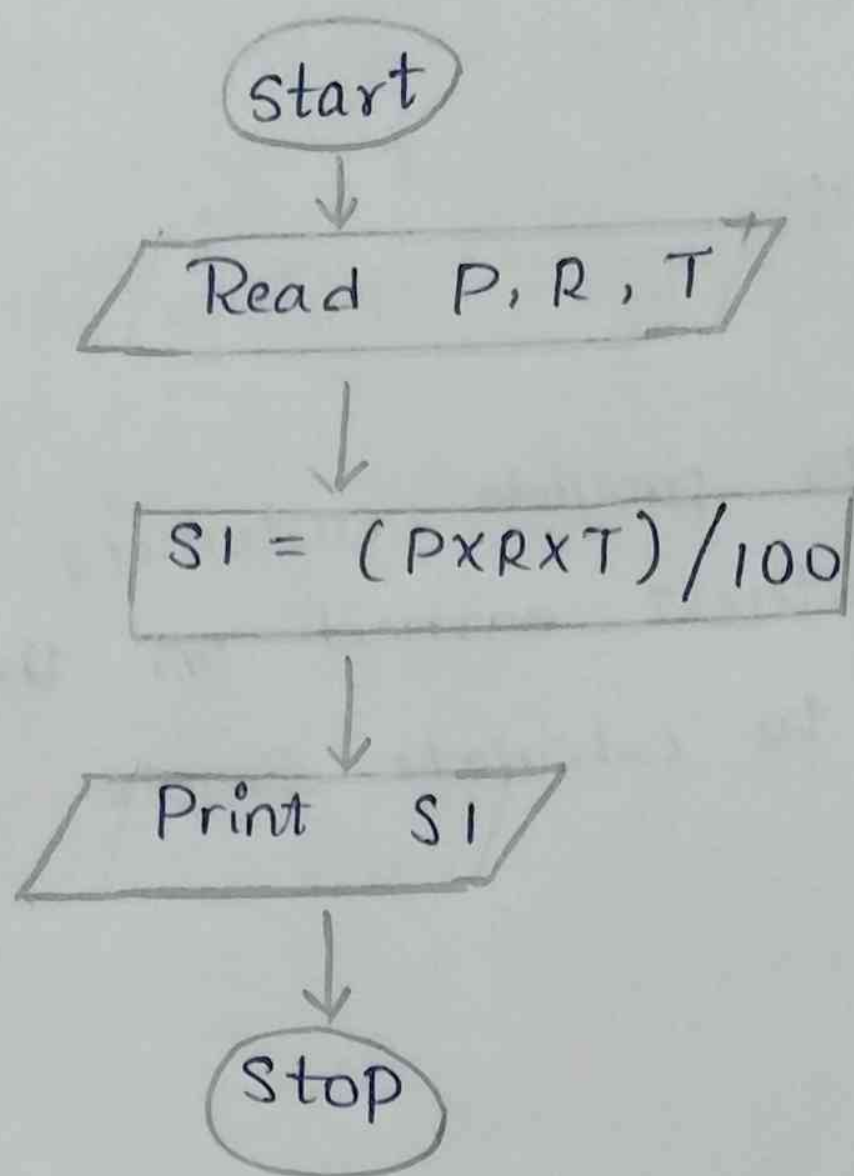
Rate (R)

Time (T)

Output:

Simple Interest

Flow chart:



5. A weather station records temperature in Celsius, but reports need to be generated in Fahrenheit. Write a C program to convert temperature from Celsius to Fahrenheit.

Pseudo Code:

Start

Read Celsius

Fahrenheit = (Celsius $\times$ 9/5) + 32

Print Fahrenheit

Stop

Input:

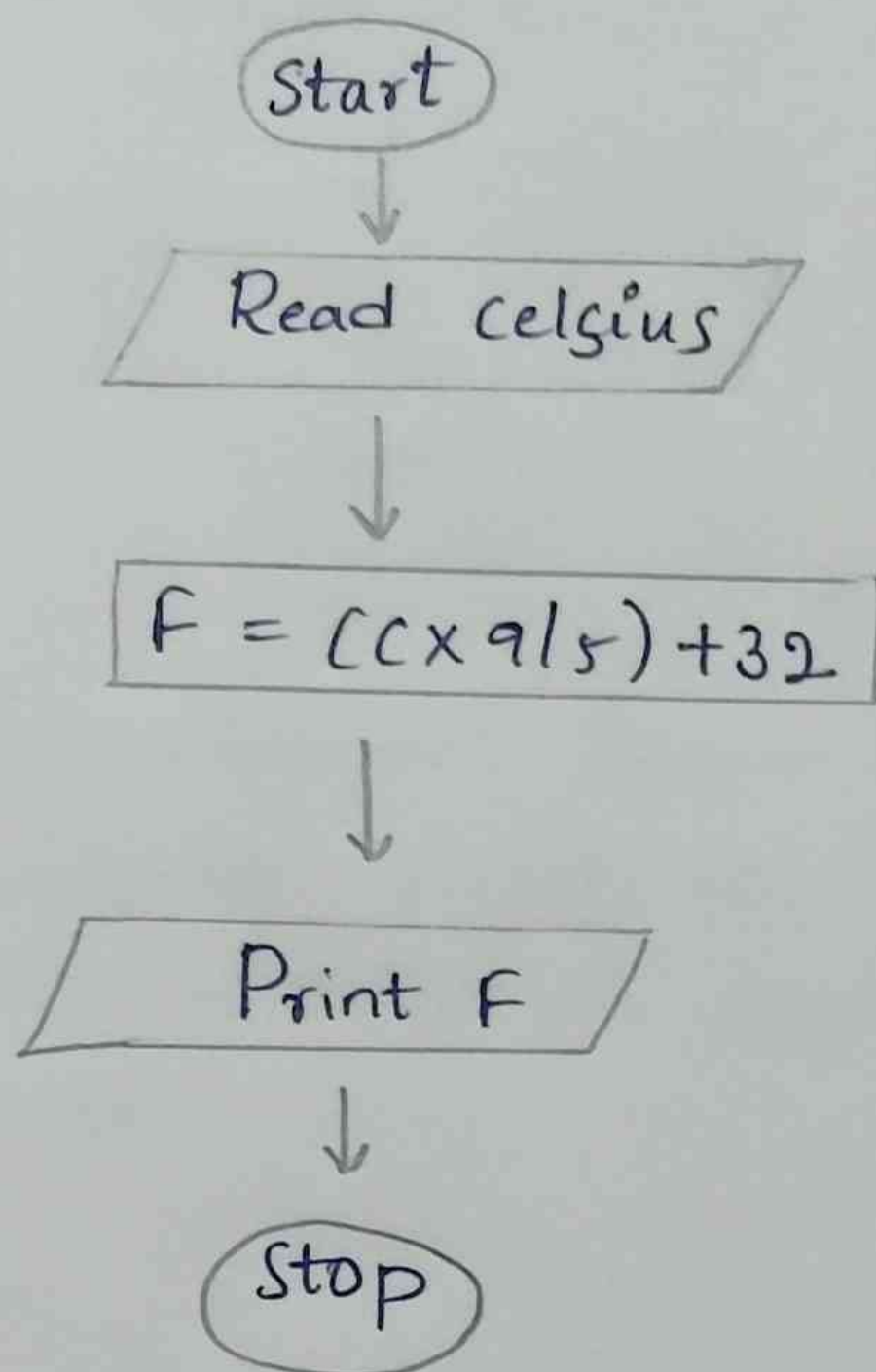
Celsius

Output:
Fahrenheit
Flow C

Output:

Fahrenheit

Flow chart:



6. A mathematics learning application need to display both the quotient and remainder when two numbers are divided. Write a C programming to find the quotient and remainder of two integers.

Pseudo code:

Start

Read A, B

Quotient = A/B

Remainder = $A \% B$

Print quotient

Print Remainder

Stop

Input:

Dividend (A)

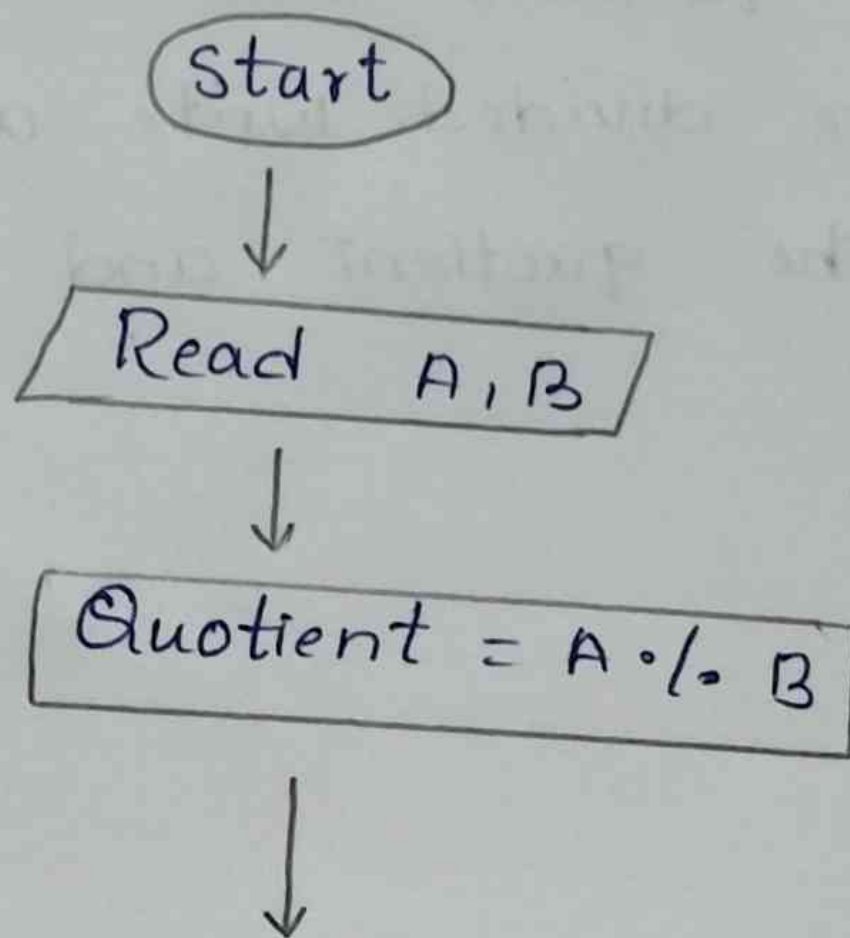
Divisor (B)

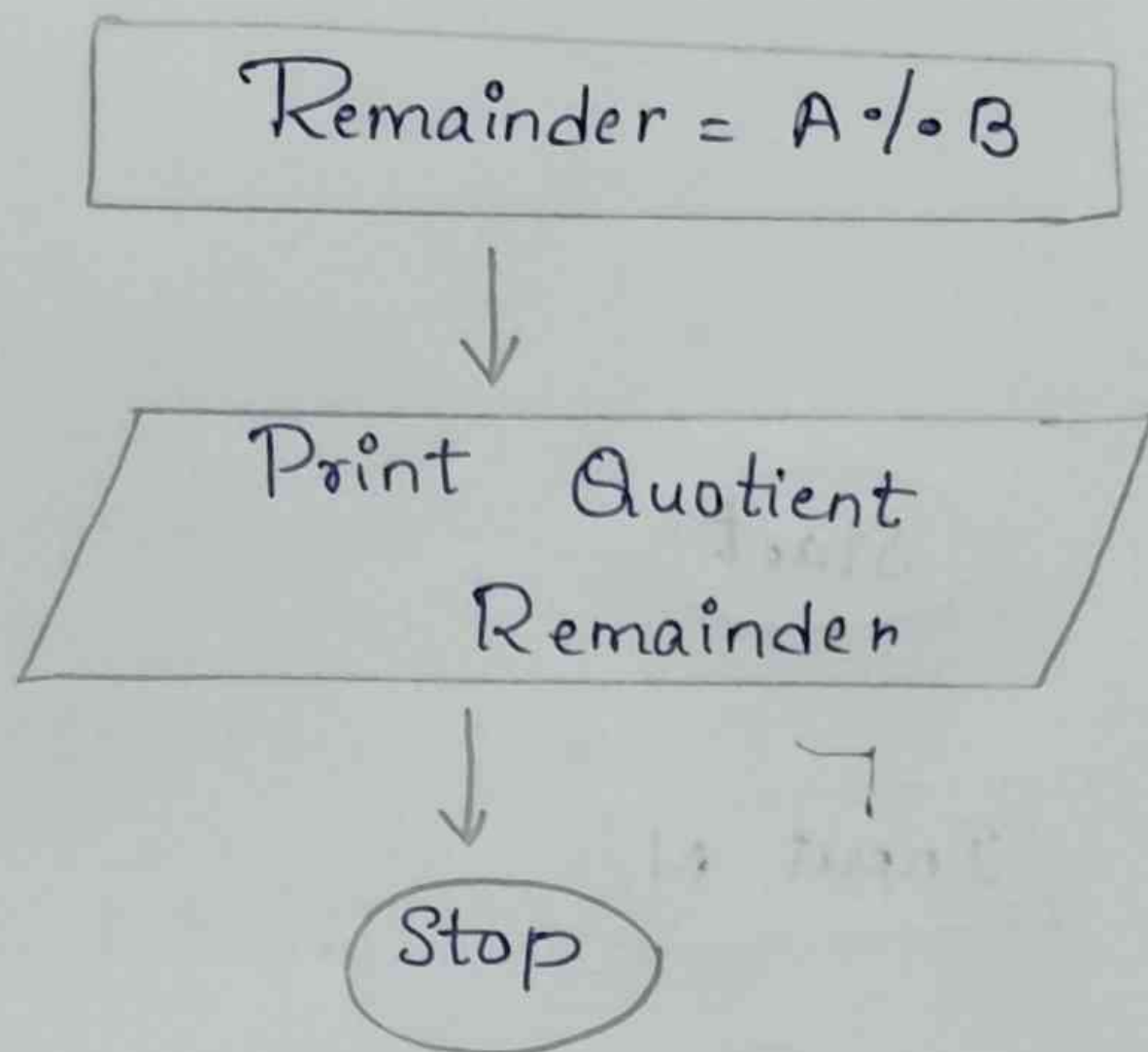
Output:

Quotient

Remainder

Flowchart:





7. An event management System allows entry based on even-numbered ticket IDs. Write a C program to determine whether a given ticket number is even or odd.

Pseudo code:

Start

Read N

If $N \% 2 == 0$

Print "Even"

Else

Print "Odd"

Stop

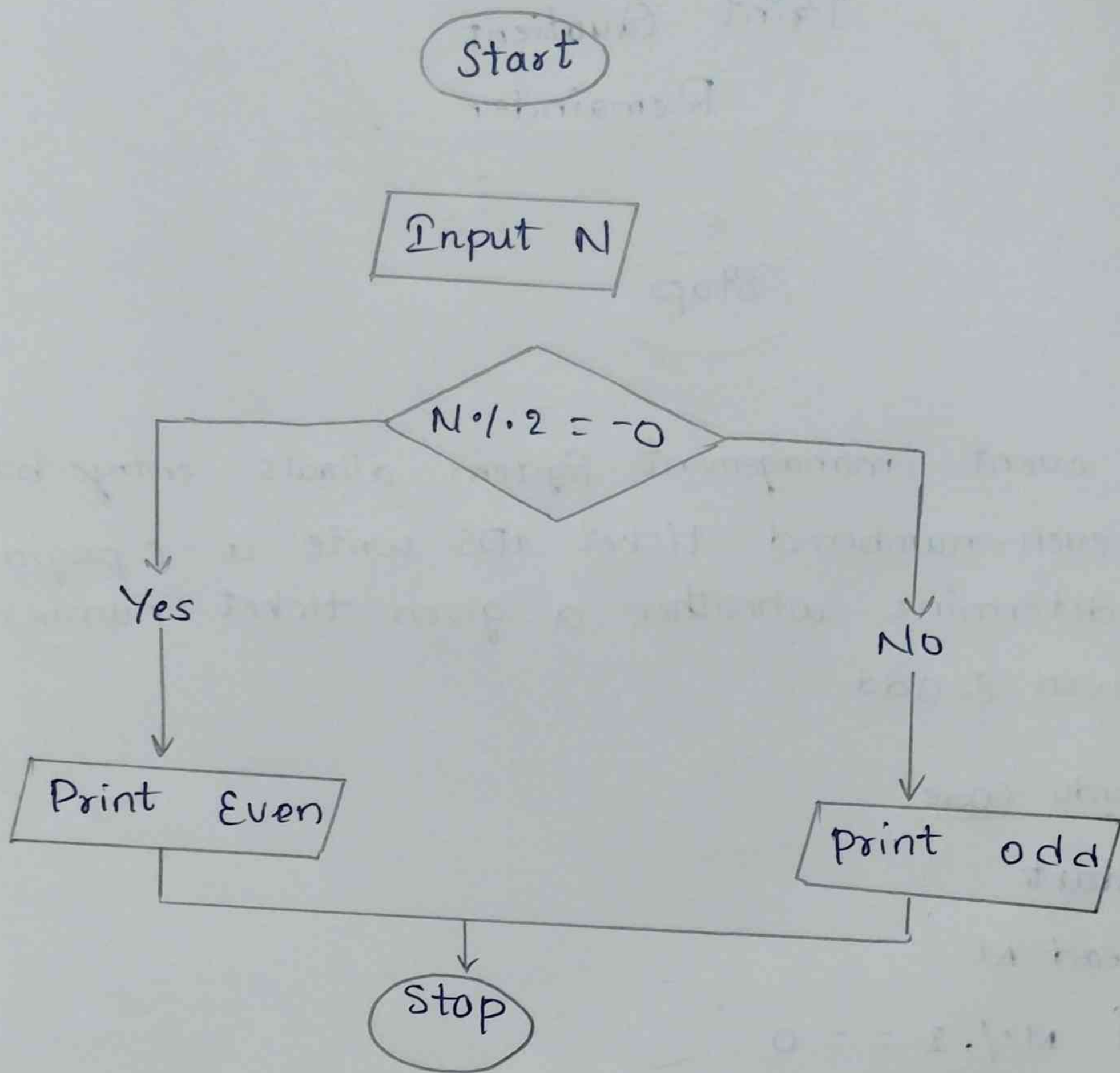
Input:

Integer N

Output:

Even or odd

Flow chart:



8. An engineer needs to solve quadratic equations of the form. Write a 'c' program to calculate and display the roots of the quadratic equations.

Pseudo Code:

Start

Read a, b, c

$$D = b * b - 4 * a * c$$

$$\text{Root 1} = (-b + \sqrt{D}) / (2 * a)$$

$$\text{Root 2} = (-b - \sqrt{D}) / (2 * a)$$

Print Root 1, Root 2

STOP

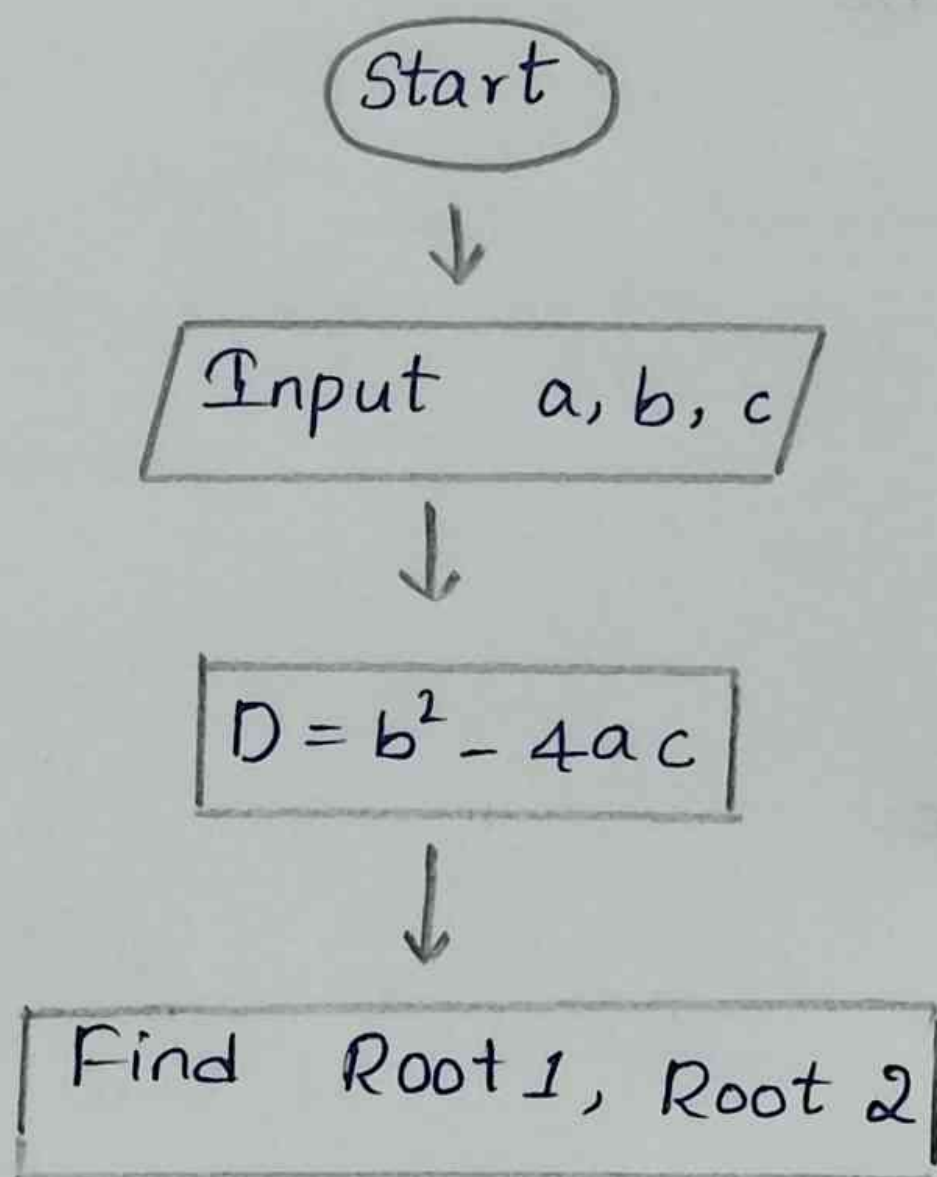
Input:

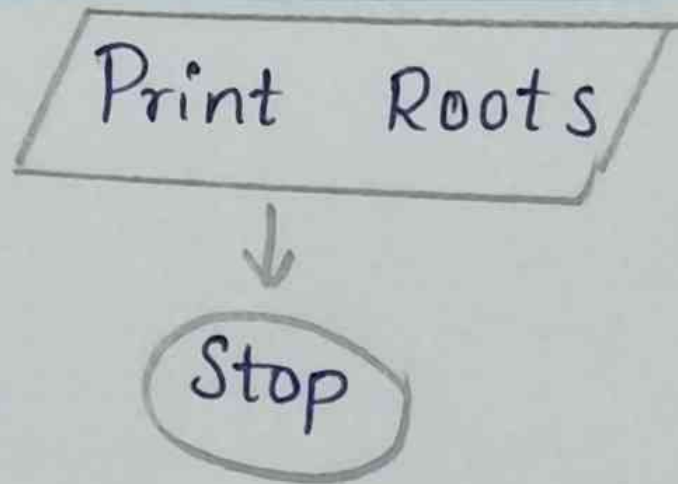
a, b, c

Output:

Root 1 and Root 2

Flow chart:





9. A scientific calculator application requires the square and cube of a given number. Write a C program to calculate the square and cube of a number entered by the user.

Pseudo Code:

Start

Read N

Square = $N * N$

Cube = $N * N * N$

Print Square, cube

stop

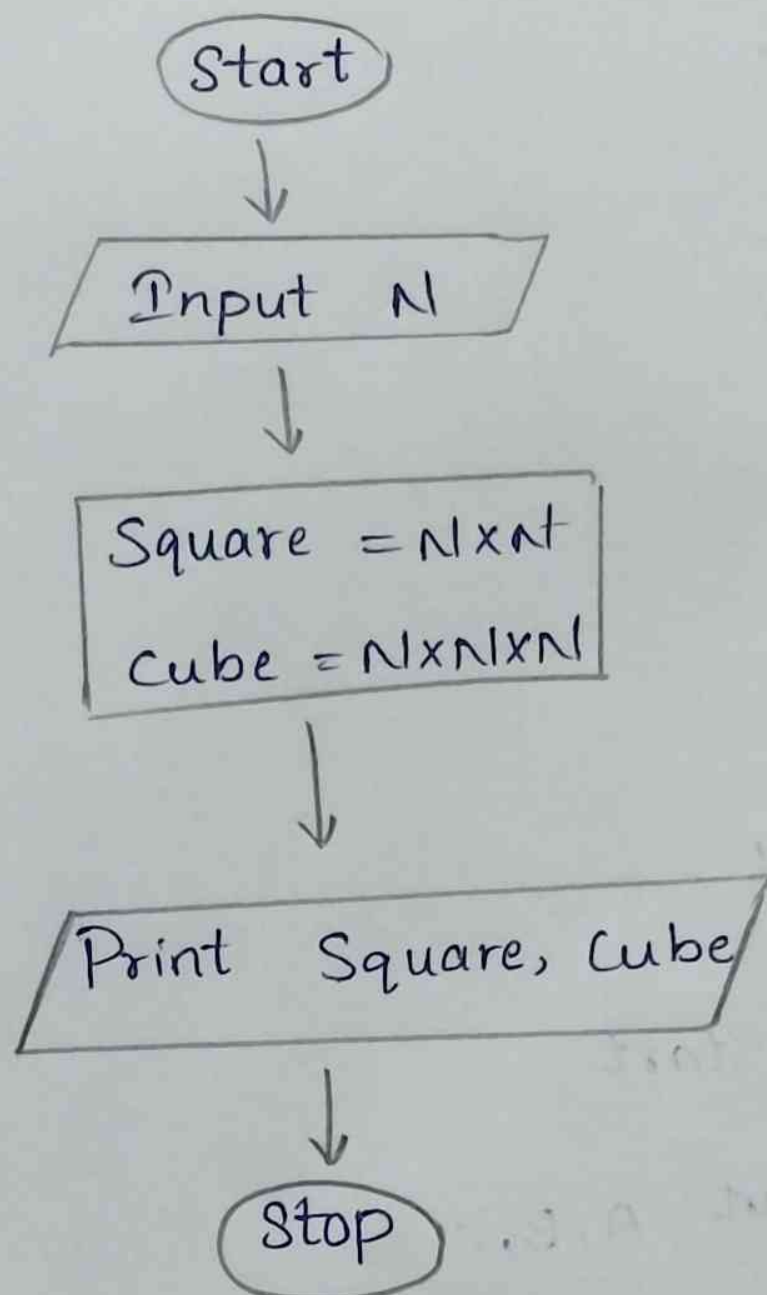
Input:

Number N

Output:

Square and cube

Flow chart :



10. A company wants to identify the top-performing sales person among three employees based on their sales figures. Write a C program to find the largest among the three numbers.

Pseudo code :

Start

Read A, B, C

If $A > B$ and $A > C$

Print A

Else if $B > A$ AND $B > C$

Print B

Else

Print C

Stop

Input :

A, B, C

Output :

largest number

Flow chart :

